Room plan: take-away games (last counter wins)

Reference only: the circle's 2–3 table already played Week 7 on October 21, 2026, before this plan was written (James, November 3). Kept for a return visit, or for a K–1 or 4–5 table that hasn't met it; the return-visit pages suit the 2–3 children who have. Week 7, Take-away games: packets for K–1, 2–3 and 4–5, and the facilitator guide, whose launch script this plan moves to the floor. This plan is unpiloted.

Why it would follow November 4

Last week's shuffle machines ran one machine again and again; this week the other side is a player who answers back. Tower cities, which the K–1 table met on October 21, asked whether a set of clues fits one city, several or none. This week's opener is a clue puzzle of that kind, and the main activity asks the same question inside a game: does my way of playing work against every reply, not just the one I expect? It is also the kind of activity this room does well. The other player enforces the rule, the counters show the state, and a game of 10 takes under a minute, so children play many games before anyone asks them to explain.

For adults: with moves of 1 or 2, the piles you want to hand your opponent are the multiples of 3 (0, 3, 6, 9, ...), and you keep it that way by answering a take of 1 with 2 and a take of 2 with 1. With moves of 1, 3 or 4 the piles to hand over are those with remainder 0 or 2 on dividing by 7 (0, 2, 7, 9, 14, 16), so Goat's pile of 7 is a first-player win in the launch and a second-player win at the midpoint. In general a pile is one to hand over exactly when every move from it reaches a pile that is not; working up from 0 sorts every pile, and for any finite list of moves the pattern ends up repeating (a pigeonhole argument, 4–5 Problem 14). Two equal piles are always ones to hand over: copy your opponent in the other pile.

Prep checklist for Wednesday afternoon

Print single-sided at 100% ("Actual size"), so the track squares stay about 0.68 inch. The guide's copy counts assume 4 children at K–1, 4 at 2–3 and 3 at 4–5; the counts below lean toward 2–3, since most of the room is there. Adjust at the door. Children get one page at a time. Prep takes about 45 minutes: 15 to print, 15 to count counters into cups, 10 for signs and cards, and the guide's 5-minute check. Nothing is heavy.

- K–1 packet, 6 pages: 4 copies, plus 4 extra copies of page 2 (six more 0–10 tracks for replays)
- Grades 2–3 packet, 5 pages: 8 copies, plus 2 extra copies of page 1 (two more 0–20 tracks)
- Grades 4–5 packet, 8 pages: 4 copies
- Facilitator guide, 10 pages: one per adult (3)
- This room plan: one per adult (3)
- Opener sheet, 2 pages, cut in half: page 1 (the main puzzle, two per page) 5 copies, page 2 (easier on top, harder below) 3 copies
- Four animal signs: DUCK, FROG, GOAT, MOUSE in big letters on four sheets of paper
- Counters of one colour, about 120: the guide's table cups (K–1 30, 2–3 30, 4–5 36; one cup per pair) and four opener cups holding 3, 5, 6 and 7. With only about 100, leave the guide's spares (6, 6 and 4) out of the table cups. A third pair at 2–3 needs another cup of 12.
- Tokens for the tracks: 8 small ones (a counter of another colour or a snap cube; a pattern block is too big)
- Light coloured pencils (22), pencils (14), blank paper (10 sheets)
- Small whiteboards and markers: 7 (for 2–3 and 4–5); one big whiteboard or easel by the floor, with 0 to 10 written in a row of big squares and a sticky note as its token
- Three early-finisher cards, handwritten (text below)
- A timer everyone can see
- The guide's five-minute check (Section 2): a token on a printed track, a pile of 10 in front of a pair, colours still readable, your table's first rule played once against yourself

Before the children arrive

- Opener half-sheets and pencils already on every table, so the first child in starts on the puzzle.

- On the floor: the four opener cups (3, 5, 6 and 7 counters, not labelled), the four animal signs face down beside them, and the big whiteboard with its 0–10 track and sticky note.
- At each table: that table's counted cups, tokens, pencils and light coloured pencils, small whiteboards at 2–3 and 4–5, and the pages face down in order with page 1 on top.
- One timer everyone can see.
- Early-finisher cards face down at each table: K–1 "Race to 10: count together from 1. On your turn say one or two numbers. Whoever says 10 wins. Who should start?"; 2–3 "Race to 21: say one, two or three numbers. Whoever says 21 wins. Can the first player always win?"; 4–5 "Race to 100: add any number from 1 to 10 to the total. Whoever reaches exactly 100 wins. What should the first player say?" (These are the guide's standing-up games. The first player wins each: say 1, then 4, 7, 10; then 5, 9, 13, 17, 21; then 12, 23, ..., 89, 100.)

Opener: whose pile? (8 minutes)

A soft logic puzzle on the tables as children arrive, in the spirit of year one's openers and of the tower cities clues, then acted out on the floor. The answer hands Goat's pile of 7 straight to the launch, which is the pile the guide's launch uses. An adult reads the clues aloud at any table that wants it.

The puzzle (most of the room). Duck, Frog, Goat and Mouse each have a pile of counters: one pile of 3, one of 5, one of 6 and one of 7 (drawn on the sheet). Write each animal's name next to its pile.

1. Frog has an odd number of counters.
2. Mouse has fewer counters than Frog.
3. Goat has more counters than Frog.
4. Duck does not have the biggest pile.

Answer: Mouse 3, Frog 5, Duck 6, Goat 7. It is the only answer, and every clue is needed (checked by computer: without clue 1 there are 3 answers, without clue 2 or 3 there are 7, without clue 4 there are 2).

Hint ladder (give one step, then walk away):

1. "Clues 2 and 3 are both about Frog. Can Frog have the biggest pile? The smallest?" (No: Mouse has fewer and Goat has more, so Frog has 5 or 6.)
2. "Which of 5 and 6 is odd?" (5, so Frog has 5. Mouse has fewer, so Mouse has 3.)
3. "Goat and Duck share 6 and 7. Which clue decides?" (Duck is not the biggest, so Goat 7 and Duck 6.)

Easier (younger children, or anyone stuck at step 1). Frog, Goat and Mouse, with piles of 3, 5 and 7. Goat has the most counters. Mouse has fewer counters than Frog. Children may write first letters. *Answer:* Mouse 3, Frog 5, Goat 7.

Harder (older children, or anyone done in two minutes). Duck, Frog, Goat, Hen and Mouse, with piles of 3, 5, 6, 7 and 8. Duck has an even number of counters. Hen has an even number of counters. Hen has more counters than Duck. Frog has exactly two more counters than Mouse. Goat has more counters than Frog. *Answer:* the main answer with Hen 8 (the even piles are 6 and 8; then Frog and Mouse are 5 and 3, since 7 and 5 would leave Goat nothing bigger than Frog). Every clue is needed.

Early finishers. "Cross out clue 4. How many answers are there now? Write a new clue that makes it exactly one again." (Two answers: Duck and Goat can swap. "Goat has more counters than Duck" fixes it, and so do others.)

On the floor (minute 6). Four children who solved it take the animal signs. Each picks up the cup that matches their answer and counts it out loud with the room, and the room checks each clue, thumbs up or down. Then Goat's cup becomes the launch's pile: Goat hands one counter to each of seven children, who stand in a line. A bridge question before the first move: "We take 1 or 2 counters each turn, and whoever takes the last counter wins. If I go first, can I make sure the last one is mine?" (Yes: take 1 and leave 6. Do not say so; play it.)

The hour

Time	Everyone	Jobs
0:00–0:08	**Opener: whose pile?** On the tables as children arrive; at minute 6, four solvers check their cups on the floor.	Solvers check sheets; adults give one hint step.
0:08–0:12	**Launch: the human pile**, with the guide's script (Section 3). Seven standing children hold Goat's 7. A taker takes 1 or 2 from the end of the line, one in each hand at most, and those children sit. The organizer goes first, takes 1, makes 3 with each take, and wins. Practise the reset signal once ("Counters down": counters on the floor, hands on heads). Then the guide's challenge: "Beat the grown-up and find the secret. Some tables change the rule."	A captain moves the whiteboard token; the rest call "allowed" or "put one back".
0:12–0:20	**Tables, round 1**: Problem 1 in every band. K–1: piles 2 to 8 (1 or 2). 2–3: piles 2, 5, 6, 7, 8 (1, 3 or 4). 4–5: piles 7, 9, 11, 12, 16 (1 or 2).	**Builder** counts and checks takes; **Chooser** picks first or second. Swap every game. 4–5: two players and a referee.
0:20–0:28	**Round 2.** K–1: Problems 2 and 3 (the adult plays one pair as a team while the other keeps playing Problem 1). 2–3: Problems 2 and 3 (the adult plays one pair while the other colours). 4–5: Problems 2 and 3.	One child says the team's move, one colours; swap.
0:28–0:33	**Floor: same pile, new rule.** Goat's 7 stand again. A 2–3 pair plays 1, 3 or 4 twice, swapping who starts; the room votes first or second before each game (second wins). Then everyone stands behind the animal pile they would rather hand over: with 1 or 2 (Mouse's 3, Duck's 6), then with 1, 3 or 4 (Goat's 7 only).	New counters; a 4–5 referee.
0:33–0:41	**Round 3.** K–1: Problems 4 and 5. 2–3: Problems 4 (Lena) and 5 (Omar). 4–5: Problems 4 and 5 (Kai).	The **Tester** tries every first move against a claim; swap.
0:41–0:48	**Round 4, by readiness.** K–1: Problem 6 or 8, or more games of 1–5. 2–3: Problem 6 or 7; 8 and 9 for the quickest. 4–5: Problem 6, then 10 and 12 with counters; 8 and 14 to explain.	Swap jobs. Early-finisher cards out.
0:48–0:55	**Floor finale: beat the grown-up.** Each table sends a pair, which names a pile (up to the children free to stand, at most 8) and its table's rule, answers the guide's sharing question ("a pile where we'd rather go second") for the whiteboard, chooses first or second, and plays the organizer on the human line.	Everyone else is a counter; a referee from another table.
0:55–1:00	Counters back in cups and counted to the guide's numbers, opener cups refilled, pages collected.	Each pair counts its own cup.

If it goes sideways

- A table finishes early: its early-finisher card. The room gets loud: reset signal, then straight to the next floor activity.
- A pair is stuck: the adult plays them once from a small pile and asks "Which pile would you rather hand your partner?", nothing more.
- Tears after a loss (the guide warns of this): short games, swap who starts, start the next game at once, and the upset child picks the next pile and who starts.
- The human line turns silly or slow: Goat's 7 in a bowl on the floor with the whiteboard track, exactly the guide's launch. The human line and the one-in-each-hand rule are untried.
- Jay and Rose: the use log has no session of Week 7, so they meet it fresh. Whoever finds a table's secret first becomes its checker, who may say "try again" but not why, and a floor captain at 0:28 and 0:48. The same goes for a child who has played Pebble Duel in the app.

How it went

(Add a dated line after the session.)